#In this code, we demonstrate the existence of the curves γ2 (r2=0) and B̃=0 in region A4 by Lemma 6.13.

$A := -(w+z)^3\left(\left(x^2y^2+z^4\right)w^4-x^2y^2z^4\right):$

$B := x^2y^5w^2z^2(w+z)^2(w-z)-x^2y^2(w+z)^3\left(w^2+z^2\right)w^2z^2(w-z)+w^4y^3z^4(w+z)^3 + w^4z^4x^3y^3:$

$R := B^2\cdot\left((w+z)^2-y^2\right)^3-A^2\cdot\left(y^2\cdot z\cdot w\right)^3$

$$R := \left(x^2y^5w^2z^2(w+z)^2(w-z)-x^2y^2(w+z)^3\left(w^2+z^2\right)w^2z^2(w-z)+w^4y^3z^4(w+z)^3 + w^4z^4x^3y^3\right)^2\left((w+z)^2-y^2\right)^3-(w+z)^6\left(\left(x^2y^2+z^4\right)w^4-x^2y^2z^4\right)^2y^6z^3w^3 \quad \textbf{(1)}$$

#Apply the rational parametrization constructed in this section.

$parametrization := \left\{ \mathrm{w}=\mathrm{a}+\frac{1}{a}, z=b+\frac{1}{b}, x=a-\frac{1}{a}+b-\frac{1}{b}, y=a-\frac{1}{a}-b+\frac{1}{b}\right\}:$

simplify(collect(numer(expand(subs(parametrization, R))), a));

$$-(a-b)^6(ab+1)^{12}\left(-2560b^{11}\left(b^2+1\right)^4\left(b^6-\frac{38}{5}b^4+\frac{42}{5}b^2-\frac{11}{5}\right)a^{21}-1536b^9\left(b^2+1\right)^4\left(b^8-\frac{43}{3}b^6+\frac{95}{3}b^4-\frac{52}{3}b^2+\frac{7}{3}\right)a^{19}+512b^7\left(b^{10}+11b^8-78b^6+81b^4-22b^2+1\right)\left(b^2+1\right)^4a^{17}+512b^7\left(b^{10}-10b^8-9b^6+42b^4-19b^2+1\right)\left(b^2+1\right)^4a^{15}-2560\left(b^8-\frac{16}{5}b^6+b^4-\frac{3}{5}b^2+\frac{3}{5}\right)b^7\left(b^2+1\right)^4a^{13}+2560b^7\left(b^2+1\right)^4\left(b^6-\frac{6}{5}b^4+\frac{8}{5}b^2-1\right)a^{11}+512b^7\left(b^2-2\right)\left(b^2+1\right)^5a^9+b^{16}+a^{32}b^{16}+\left(72b^{20}-76b^{18}+1288b^{16}+692b^{14}-696b^{12}-512b^{10}\right)a^6+\left(6b^{20}-32b^{18}+44b^{16}-288b^{14}-250b^{12}\right)a^4+\left(-4b^{18}+8b^{16}-4b^{14}\right)a^2+\left(-4b^{18}+8b^{16}-4b^{14}\right)a^{30}+\left(6b^{20}-32b^{18}+44b^{16}-32b^{14}+6b^{12}\right)a^{28}+\left(-1024b^{22}+72b^{20}+692b^{18}+8b^{16}-76b^{14}\right. \quad \textbf{(2)}$$

$$+ 72 b^{12}\big) a^{26} + \big(-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18}$$

$$- 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^{8} + 72 b^{6} + 2 b^{4}\big) a^{20} + \big(b^{32}$$

$$+ 8 b^{30} + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16}$$

$$- 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^{8} - 120 b^{6} + 20 b^{4} + 8 b^{2} + 1\big) a^{16} + \big($$

$$-9 b^{24} - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^{8}\big)$$

$$a^{8} - 9 \left(b^{16} + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^{8} - 72 b^{6} - 16 b^{4} + 8 b^{2} + 1\right) b^{8} a^{24}$$

$$+ 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} + \frac{2163}{2} b^{8} - 159 b^{6} - 27 b^{4}\right.$$

$$\left. + b^{2} + 1\right) b^{6} a^{22} - 4 b^{2} \big(b^{28} + 76 b^{26} - 539 b^{24} - 2118 b^{22} + 5211 b^{20} + 36360 b^{18}$$

$$+ 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^{8} - 518 b^{6} + 37 b^{4} + 12 b^{2} + 1\big)$$

$$a^{18} - 1024 b^{9} \big(b^{2} + 1\big)^{4} a^{7} + 8 b^{6} \left(b^{20} + b^{18} - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10}\right.$$

$$\left. - \frac{9357}{2} b^{8} - 3231 b^{6} - 2747 b^{4} - 1055 b^{2} - 127\right) a^{10} + 2 b^{4} \big(b^{24} + 36 b^{22} + 180 b^{20}$$

$$- 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} - 27212 b^{8} - 6620 b^{6} - 2636 b^{4}$$

$$+ 36 b^{2} + 1\big) a^{12} - 4 b^{2} \big(b^{28} + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18}$$

$$+ 41514 b^{16} + 57358 b^{14} + 65514 b^{12} + 34120 b^{10} + 4315 b^{8} + 1018 b^{6} - 155 b^{4} + 12 b^{2}$$

$$\left. + 1\big) a^{14} - 1024 \big(b^{2} - 5\big) \left(b^{2} - \frac{1}{2}\right) b^{13} \big(b^{2} + 1\big)^{4} a^{23}\right) (a + b)^{6} \big(b^{2} + 1\big)^{3} \big(a^{2} + 1\big)^{3}$$

#Define r2

$$r2 := \left(b^{16} + a^{32} b^{16} - 1024 b^{9} \big(b^{2} + 1\big)^{4} a^{7} + 8 \left(b^{20} + b^{18} - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12}\right.\right.$$

$$\left. - 3875 b^{10} - \frac{9357}{2} b^{8} - 3231 b^{6} - 2747 b^{4} - 1055 b^{2} - 127\right) b^{6} a^{10} + 2 b^{4} \big(b^{24} + 36 b^{22}$$

$$+ 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} - 27212 b^{8} - 6620 b^{6}$$

$$
\begin{aligned}
&- 2636\, b^4 + 36\, b^2 + 1 \big)\, a^{12} - 4\, b^2\, \big(b^{28} + 12\, b^{26} + 37\, b^{24} + 378\, b^{22} + 4187\, b^{20} + 19656\, b^{18} \\
&+ 41514\, b^{16} + 57358\, b^{14} + 65514\, b^{12} + 34120\, b^{10} + 4315\, b^8 + 1018\, b^6 - 155\, b^4 + 12\, b^2 + 1 \big) \\
&a^{14} - 4\, b^2\, \big(b^{28} + 76\, b^{26} - 539\, b^{24} - 2118\, b^{22} + 5211\, b^{20} + 36360\, b^{18} + 92394\, b^{16} + 78414\, b^{14} \\
&+ 38058\, b^{12} + 17992\, b^{10} - 2085\, b^8 - 518\, b^6 + 37\, b^4 + 12\, b^2 + 1 \big)\, a^{18} + 8\, \Big(b^{20} + b^{18} - 699\, b^{16} \\
&+ 2369\, b^{14} - \frac{1037}{2}\, b^{12} - 1827\, b^{10} + \frac{2163}{2}\, b^8 - 159\, b^6 - 27\, b^4 + b^2 + 1 \Big)\, b^6\, a^{22} - 9\, \Big(b^{16} \\
&+ \frac{4168}{9}\, b^{14} - 528\, b^{12} - 328\, b^{10} + \frac{3698}{9}\, b^8 - 72\, b^6 - 16\, b^4 + 8\, b^2 + 1 \Big)\, b^8\, a^{24} + \big(-4\, b^{18} \\
&+ 8\, b^{16} - 4\, b^{14} \big)\, a^{30} + \big(6\, b^{20} - 32\, b^{18} + 44\, b^{16} - 32\, b^{14} + 6\, b^{12} \big)\, a^{28} + \big(-1024\, b^{22} + 72\, b^{20} \\
&+ 692\, b^{18} + 8\, b^{16} - 76\, b^{14} + 72\, b^{12} \big)\, a^{26} + \big(-254\, b^{28} + 840\, b^{26} + 6504\, b^{24} - 696\, b^{22} \\
&- 25752\, b^{20} - 169584\, b^{18} - 181788\, b^{16} - 85104\, b^{14} - 50328\, b^{12} - 4792\, b^{10} + 360\, b^8 + 72\, b^6 \\
&+ 2\, b^4 \big)\, a^{20} + \big(b^{32} + 8\, b^{30} + 20\, b^{28} + 1672\, b^{26} + 1950\, b^{24} - 32376\, b^{22} - 157780\, b^{20} \\
&- 317080\, b^{18} - 279672\, b^{16} - 222104\, b^{14} - 125524\, b^{12} - 7288\, b^{10} - 3426\, b^8 - 120\, b^6 + 20\, b^4 \\
&+ 8\, b^2 + 1 \big)\, a^{16} + \big(-9\, b^{24} - 72\, b^{22} + 144\, b^{20} - 1912\, b^{18} + 142\, b^{16} + 4232\, b^{14} + 2192\, b^{12} \\
&- 1864\, b^{10} - 265\, b^8 \big)\, a^8 + \big(72\, b^{20} - 76\, b^{18} + 1288\, b^{16} + 692\, b^{14} - 696\, b^{12} - 512\, b^{10} \big)\, a^6 \\
&+ \big(6\, b^{20} - 32\, b^{18} + 44\, b^{16} - 288\, b^{14} - 250\, b^{12} \big)\, a^4 + \big(-4\, b^{18} + 8\, b^{16} - 4\, b^{14} \big)\, a^2 - 1024\, \big(b^2 \\
&- 5 \big)\, \Big(b^2 - \frac{1}{2} \Big)\, \big(b^2 + 1 \big)^4\, b^{13}\, a^{23} + 2560\, \Big(b^6 - \frac{6}{5}\, b^4 + \frac{8}{5}\, b^2 - 1 \Big)\, \big(b^2 + 1 \big)^4\, b^7\, a^{11} \\
&+ 512\, b^7\, \big(b^2 - 2 \big)\, \big(b^2 + 1 \big)^5\, a^9 + 512\, b^7\, \big(b^{10} - 10\, b^8 - 9\, b^6 + 42\, b^4 - 19\, b^2 + 1 \big)\, \big(b^2 \\
&+ 1 \big)^4\, a^{15} - 2560\, \big(b^2 + 1 \big)^4\, \Big(b^8 - \frac{16}{5}\, b^6 + b^4 - \frac{3}{5}\, b^2 + \frac{3}{5} \Big)\, b^7\, a^{13} + 512\, b^7\, \big(b^{10} + 11\, b^8 \\
&- 78\, b^6 + 81\, b^4 - 22\, b^2 + 1 \big)\, \big(b^2 + 1 \big)^4\, a^{17} - 2560\, \Big(b^6 - \frac{38}{5}\, b^4 + \frac{42}{5}\, b^2 - \frac{11}{5} \Big)\, \big(b^2 \\
&+ 1 \big)^4\, b^{11}\, a^{21} - 1536\, \Big(b^8 - \frac{43}{3}\, b^6 + \frac{95}{3}\, b^4 - \frac{52}{3}\, b^2 + \frac{7}{3} \Big)\, \big(b^2 + 1 \big)^4\, b^9\, a^{19} \Big) :
\end{aligned}
$$

#Define $\tilde{B}$

$$
\begin{aligned}
\tilde{B} := &\bigg(\Big(-\frac{1}{2}\, b^5 + b^7 \Big)\, a^{10} + \Big(\frac{3}{2}\, b^7 - 3\, b^5 + b^3 \Big)\, a^8 + \frac{b^2\, \big(b^2 - 5 \big)\, \big(b^2 + 1 \big)^3\, a^7}{2} + \Big(-\frac{1}{2}\, b \\
&- \frac{7}{2}\, b^5 + 3\, b^3 \Big)\, a^6 + \Big(\frac{1}{2} - \frac{3}{2}\, b^8 - 4\, b^6 - 3\, b^4 \Big)\, a^5 + \Big(\frac{5}{2}\, b^3 - b - \frac{1}{2}\, b^7 \Big)\, a^4 - \big(b^2
\end{aligned}
$$

$+ 1)^3 a^3 + \left(b^5 - \frac{1}{2} b\right) a^2 - \frac{b^3}{2}\right):$

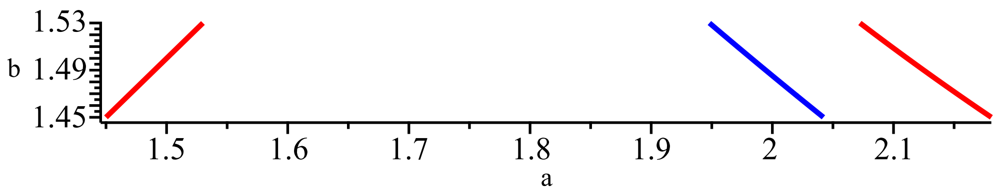
#Plot the curves ñ=0 and p̃=0 in region A4.

$r := a - b:$

with(plots) :

implicitplot([r2 = 0, B̃ = 0, r = 0], a = 1 .. 2.2, b = 1.45 .. 1.53,
scaling = constrained,
gridrefine = 2,
thickness = 2,
color = [blue, red, red],
labels = ["a", "b"],
title = "Graphic for r2=0 and B̃=0 in A4");

Graphic for r2=0 and B̃=0 in A4



#Apply the Möbius transformation on the rectangle A4 = [sqrt(3), 2.2) × [1.45, 1.53) *constructing r2*

$$moebius := \left\{ a = \frac{\mathrm{sqrt}(3) + \left(\frac{11}{5}\right) c}{1 + c}, \right.$$

$$b = \frac{\frac{29}{20} + \left(\frac{153}{100}\right) d}{1 + d}\right\} :$$

$r2 := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{moebius}, r2))), c) :$

#Compute the degree of r2 in c

degree(r2, c);

32 **(3)**

#Define the coefficients ci = coeff(r2, c, i)
seq(evalf(assign(cat('c', i), coeff(r2, c, i))), i = 0 .. 32);

#Compute the all degree of ci in d
seq(degree(eval(cat('c', i)), d), i = 0 .. 32);

32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32 **(4)**

#From here we will analyze the signal variation of all coefficients ci

seq(sign(coeff(c32, d, i)), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(5)**

seq(sign(evalf(coeff(c31, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(6)**

seq(sign(evalf(coeff(c30, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(7)**

seq(sign(evalf(coeff(c29, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(8)**

seq(sign(evalf(coeff(c28, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(9)**

seq(sign(evalf(coeff(c27, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(10)**

seq(sign(evalf(coeff(c26, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(11)**

seq(sign(evalf(coeff(c25, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(12)**

seq(sign(evalf(coeff(c24, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(13)**

seq(sign(evalf(coeff(c23, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(14)**

seq(sign(evalf(coeff(c22, d, i))), i = 32 .. 0, −1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 **(15)**

seq(sign(evalf(coeff(c21, d, i))), i = 32 .. 0, −1);
fsolve(c21 = 0, d, 0 .. infinity);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, −1, −1, −1, −1, −1

0.1740942732 **(16)**

seq(sign(evalf(coeff(c20, d, i))), i = 32 .. 0, −1);
fsolve(c20 = 0, d, 0 .. infinity);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1

0.4386637085 **(17)**

seq(sign(evalf(coeff(c19, d, i))), i = 32 .. 0, −1);
fsolve(c19 = 0, d, 0 .. infinity);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1

0.8585518125 **(18)**

seq(sign(evalf(coeff(c18, d, i))), i = 32 .. 0, −1);
fsolve(c18 = 0, d, 0 .. infinity);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1

1.627500653 **(19)**

seq(sign(evalf(coeff(c17, d, i))), i = 32 .. 0, −1);
fsolve(c17 = 0, d, 0 .. infinity);

1, 1, 1, 1, 1, 1, 1, 1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1

3.490693424 **(20)**

seq(sign(evalf(coeff(c16, d, i))), i = 32 .. 0, −1);
fsolve(c16 = 0, d, 0 .. infinity);

1, 1, 1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1

14.54689523 **(21)**

seq(sign(evalf(coeff(c15, d, i))), i = 32 .. 0, −1);

−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1 **(22)**

seq(sign(evalf(coeff(c14, d, i))), i = 32 .. 0, −1);

−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1 **(23)**

seq(sign(evalf(coeff(c13, d, i))), i = 32 .. 0, −1);

−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1 **(24)**

seq(sign(evalf(coeff(c12, d, i))), i = 32 .. 0, −1);

−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, **(25)**

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$seq(sign(evalf(coeff(c11, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{26}$$

$seq(sign(evalf(coeff(c10, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{27}$$

$seq(sign(evalf(coeff(c9, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{28}$$

$seq(sign(evalf(coeff(c8, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{29}$$

$seq(sign(evalf(coeff(c7, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{30}$$

$seq(sign(evalf(coeff(c6, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{31}$$

$seq(sign(evalf(coeff(c5, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{32}$$

$seq(sign(evalf(coeff(c4, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{33}$$

$seq(sign(evalf(coeff(c3, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{34}$$

$seq(sign(evalf(coeff(c2, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{35}$$

$seq(sign(evalf(coeff(c1, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{36}$$

$seq(sign(evalf(coeff(c0, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{37}$$

#*Apply the Möbius transformation on the rectangle [sqrt(3), 2)* $\times$ $[1.53, 1.6)$ *constructing* $\check{B}$
$\check{B} := \text{collect}(\text{numer}(\text{expand}(\text{subs}(moebius, \tilde{B}))), d)$:

#*Compute the degree of* $\check{B}$ *in c*

$degree(\check{B}, c);$

$$10 \tag{38}$$

#*Define the coefficients Bi = coeff* $(\check{B}, c, i)$
$seq(evalf(assign(cat('B', i), coeff(\check{B}, c, i))), i = 0 .. 10);$

#*Compute the all degree of Bi in d*
$seq(degree(eval(cat('B', i)), d), i = 0 .. 10);$

$$10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10 \tag{39}$$

#*From here we will analyze the signal variation of all coefficients Bi*

$seq(sign(coeff(B10, d, i)), i = 10 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{40}$$

$seq(sign(evalf(coeff(B9, d, i))), i = 10 .. 0, -1);$
$fsolve(B9 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1$$

$$0.4809805182 \tag{41}$$

$seq(sign(evalf(coeff(B8, d, i))), i = 10 .. 0, -1);$
$fsolve(B8 = 0, d, 0 .. \text{infinity});$

$$1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$4.666651578 \tag{42}$$

$seq(sign(evalf(coeff(B7, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{43}$$

$seq(sign(evalf(coeff(B6, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{44}$$

$seq(sign(evalf(coeff(B5, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{45}$$

$seq(sign(evalf(coeff(B4, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{46}$$

$seq(sign(evalf(coeff(B3, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{47}$$

$seq(sign(evalf(coeff(B2, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{48}$$

$seq(sign(evalf(coeff(B1, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{49}$$

$seq(sign(evalf(coeff(B0, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{50}$$